

Module 2

1. What is Exploratory Testing?

Ans Exploratory testing is a software testing approach in which testers actively learn about an application, design tests, and execute them at the same time, rather than following a detailed pre-written test script.

2. What is traceability matrix?

Ans A Traceability Matrix is a document used in software testing and project management to ensure that all requirements are covered by test cases and properly validated.

3. What is Boundary value testing?

Ans Boundary Value Testing is a software testing technique used to test values at the edges or boundaries of input ranges.

4. What is Equivalence partitioning testing?

Ans Equivalence Partitioning is a black-box testing technique where input data is divided into groups (partitions) that are expected to behave similarly.

5. What is Integration testing?

Ans Integration Testing verifies whether different modules/components of an application work correctly together.

6. What determines the level of risk?

Ans Risk level is determined by the probability of failure and the impact it can cause. Higher impact and higher chances of occurrence mean higher risk.

7. What is Alpha testing?

Ans Alpha testing is performed by internal testers or developers before releasing the software to users. It helps identify defects in a controlled environment.

8. What is beta testing?

Ans Beta testing is performed by real users in a real environment before final release. It helps gather feedback and detect remaining issues.

9. What is component testing?

Ans Component testing verifies individual modules or components separately from the entire system. It ensures each unit works correctly.

10. What is functional system testing?

Ans Functional system testing validates the complete system against functional requirements. It checks whether all features work as expected.

11. What is Non-Functional Testing?

Ans Non-functional testing checks system attributes such as performance, security, usability, and reliability. It focuses on how well the system performs.

12. What is GUI Testing?

Ans GUI testing verifies the graphical user interface of an application. It checks elements like buttons, menus, fields, and layout for proper functionality and appearance.

13. What is Ad hoc testing?

Ans Ad hoc testing is an informal testing method performed without predefined test cases. It relies on tester experience and intuition to find defects.

14. What is load testing?

Ans Load testing evaluates system performance under expected user load. It checks response time, stability, and system behavior.

15. What is stress Testing?

Ans Stress testing checks how the system behaves under extreme or beyond-normal load conditions. It helps identify the breaking point of the system.

16. What is white box testing and list the types of white box testing?

Ans White box testing tests the internal code structure, logic, and paths of an application. It is usually performed by developers.

Types:

- Statement Coverage
- Branch Coverage
- Path Coverage
- Loop Testing

17. What is black box testing? What are the different black box testing techniques?

Ans Black box testing validates software functionality without knowing internal code structure. It focuses on input and output behavior.

Techniques:

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing

18. Mention what are the categories of defects?

Ans Categories of defects are:

- a. Critical Defect
- b. Major Defect
- c. Minor Defect
- d. Trivial Defect

19. Mention what bigbang testing is?

Ans Big Bang testing is an integration testing approach where all modules are combined and tested together at once. It is mainly used for small systems.

20. What is the purpose of exit criteria?

Ans Purpose of exit criteria is to define when testing is complete and the software is ready for release.

21. When should "Regression Testing" be performed?

Ans Regression testing is performed after code changes, bug fixes, or enhancements. It ensures existing functionality is not affected.

22. What are 7 key principles? Explain in detail?

Ans

The 7 key principles of software testing are:

- Testing shows Presence of defects, not absence.
- Exhaustive testing is impossible.
- Early testing saves time and cost.
- Defects are usually found in a few modules.
- Repeating same tests becomes less effective.
- Testing depends on the project type.
- Error-free Software may still fail user needs.

23. Difference between QA v/s QC v/s Tester

QA (Quality Assurance)	QC (Quality Control)	Tester
Focuses on the improving process	Focuses on checking the product	Tests the software for bugs
Prevents defects	Finds defects	Executes test cases
Process-oriented	Product-oriented	Part of QC process

24. Difference between Smoke and Sanity?

Smoke Testing	Sanity Testing
Checks basic functions of software	Checks specific changed features
Don after new build release	Done after bug fixes or changes
Wide and basic testing	Small and focused testing

25. Difference between verification and Validation

Verification	Validation
Checks whether software is built correctly	Checks whether correct software is built

Done during development	Done after development
Focuses on process	Focuses on final product

26.Explain types of Performance testing.

Ans

- Load Testing – check software performance with many users.
- Stress Testing – Check software under heavy load.
- Spike Testing – Check performance with sudden increase in user.
- Endurance Testing – Check software performance for a long time.
- Volume Testing – Check performance with large amount of data.

27.What is Error, Defect, Bug and failure?

Ans

- Error – A mistake made by a developer or programmer.
- Defect – A problem found in the software.
- Bug – Another name for a defect in software.
- Failure – when software does not work as expected.

28.Difference between Priority and Severity

Priority	Severity
Shows how quickly a bug should be fixed	Shows how serious the bug is
Decided by manager or tester	Decided by tester
Related to business impact	Related to system impact

29.What is Bug Life Cycle?

Ans Bug life cycle is the process that a bug follows from finding the bug until it is fixed and close.

30.Explain the difference between Functional testing and Non-Functional testing.

Functional Testing	Non-Functional Testing
Checks software functions and features	Checks performance, security, speed, etc.
Tests what the software does	Tests how the software works
Example: Login testing	Example: Load testing

31.What is the difference between the STLC (Software Testing Life Cycle) and SDLC(Software Development Life Cycle)?

STLC	SDLC
Focuses on software testing	Focuses on software development
Include test planning, testing, and bug reporting	Include test planning, design, coding, testing, and maintenance
Done by testing team	Done by development and testing teams

32.What is the difference between test scenarios, test cases, and test script?

Test Scenarios	Test Case	Test Script
Describes what to test	Describe steps and expected result	Contains automated testing code
High- level idea	Detailed testing steps	Used for automation testing
Example: Login feature	Example: Enter username and Password	Example: Selenium script

33.Explain what Test Plan is? What is the information that should be covered.

Ans A Test plan is a document that describe how testing will be done in a project.

1. Testing objectives
2. Scope of testing
3. Test Strategy
4. Resources and roles
5. Test Schedule
6. Tools used
7. Risks and reports

34.What is priority?

Ans Priority shows how quickly a bug or issue should be fixed.

35.What is severity?

Ans Severity shows how serious a bug is and how much it affects the software.

36. Bug categories are...

Ans

1. Critical
2. Major
3. Minor
4. Trivial

37. Advantage of Bugzilla .

Ans

- a. Easy bug tracking and management.
- b. Helps assigning bugs to team members.
- c. Tracks bug status and progress.
- d. Generates reports for testing activities.
- e. Free and open-source tool.

38. Difference between priority and severity

Priority	Severity
Shows how quickly a bug should be fixed	Shows how serious the bug is
Related to business impact	Related to software impact
Set by Tester/Manager	Usually set by Tester

39. What are the different Methodologies in Agile Development Model?

- Scrum – Work is done in small time periods called sprints.
- Kanban – Uses a board to track work progress.
- Extreme programming (xp) – Focuses on coding quality and customer feedback.
- Lean – Reduces waste and improves efficiency.
- Crystal – Flexible method based on team size and project needs.

40. Explain the difference between Authorization and Authentication in Web testing. What are the common problems faced in Web testing?

Authentication	Authorization
Checks who the user is	Checks what the user can access
Example: Login with username and password	Example: Admin can delete data, user cannot

Common Problems in Web Testing

- Broken links
- Slow page loading
- Browser compatibility issues
- Login and Security Problems
- Form validation errors

41. When to used Usablity Testing?

Ans Usability Testing is used to check whether an application is easy to use, user-friendly, and understandable for end users. It helps identify problems in navigation, design, functionality, and user experience.

42.What is the procedure for GUI Testing?

Ans GUI (Graphical User Interface) Testing verifies the visual elements and behavior of the application interface.